

Week 6 — For Loops & Patterns

Name: ____ Date: ____

range chant: **starts at 0 · jumps by step · stops BEFORE the wall.**

Today's words

Word	What it means
for loop	repeats a block a set number of times
loop variable	the lap counter (<code>i</code>) — changes every lap
<code>range(start, stop, step)</code>	deals numbers: from start, jumping by step, stopping <i>before</i> stop
accumulator	a snowball variable: <code>total = total + n</code>
<code>"*" * i</code>	<code>i</code> copies of the star, glued together

1 · Match the sequence to its range

Draw a line from each sequence to the `range` that makes it. (One range is a fake — it matches nothing!)

Sequence		range call
0, 1, 2, 3, 4		<code>range(2, 11, 2)</code>
2, 4, 6, 8, 10		<code>range(5)</code>
5, 10, 15, 20		<code>range(10, 0, -1)</code>
10, 9, 8, ... 1		<code>range(1, 5)</code>
		<code>range(5, 21, 5)</code>

2 · Be the computer: trace the snowball

```
total = 0
for n in range(1, 5):
    total = total + n
print(total)
```

Fill in the table, one lap per row, then the final line:

lap	<code>n</code>	total after the lap
1		
2		
3		
4		

The screen shows: _____

3 · What prints? ★

```
for i in range(1, 4):
    print("*" * i)
```

Draw the exact output:

Brain teaser (optional — take it home)

Gauss beat his teacher by pairing numbers: $1+100, 2+99, \dots = 50$ pairs of $101 = \mathbf{5050}$.

Use his trick — no computer, no calculator — to add $1 + 2 + 3 + \dots + 1000$. How many pairs? What does each pair add up to? Then the showdown question: which range would make a loop check your answer — `range(1000)` or `range(1, 1001)` — and *why does the other one get it wrong?* Bring your answer next week for a shout-out.